TECHNICAL PRODUCT PITCH

SWARM

An Ecosystem Strategy Game Built on Pheromone Physics

Moving beyond unit micro-management: Shaping emergent behavior through invisible information fields.

Pikmin | Noita | Oxygen Not Included | Rain World

DESIGN PHILOSOPHY

Core Vision

"Players feel like they are bending nature rather than commanding it. You don't build armies—you shape information."

— SWARM CORE VALUE PROPOSITION

THE PROBLEM

The RTS Optimization Trap

Modern RTS Dilemma

Traditional strategy games devolve into rigid optimization loops:

Build optimal units → Spam meta strategy → Memorize execution.

The human player effectively acts like a deterministic computer, removing organic joy and unexpected outcomes.

Emergent Storytelling

Players crave games where stories emerge naturally from basic underlying rules:

`RimWorld` `Dwarf Fortress` `Factorio` `Noita`

Swarm brings this systemic magic to ecosystem macro-strategy.

THE SOLUTION

Pheromone Vector Simulation

Information Control

Players never click an ant or command an army. You lay down invisible vector fields: Hunt, Nest, Fear, and Food signals.

Emergent Behavior

Every individual agent evaluates local environmental parameters independently. Micro decisions coalesce into macro swarm intelligence.

Dynamic Diffusion

Smell diffuses along terrain contours, carries across wind vectors, dilutes in rain, and burns away in forest fires.

GAMEPLAY LOOP

Systemic Progression Architecture

1. Explore

Probe procedural biomes. Locate food sources and identify apex environmental predators.

2. Manipulate

Synthesize chemical pheromone trails to subtly guide swarm expansion and worker paths.

3. Adapt

Survive dynamic floods, rain, and competing ant colonies through ecosystem leverage.

4. Evolve

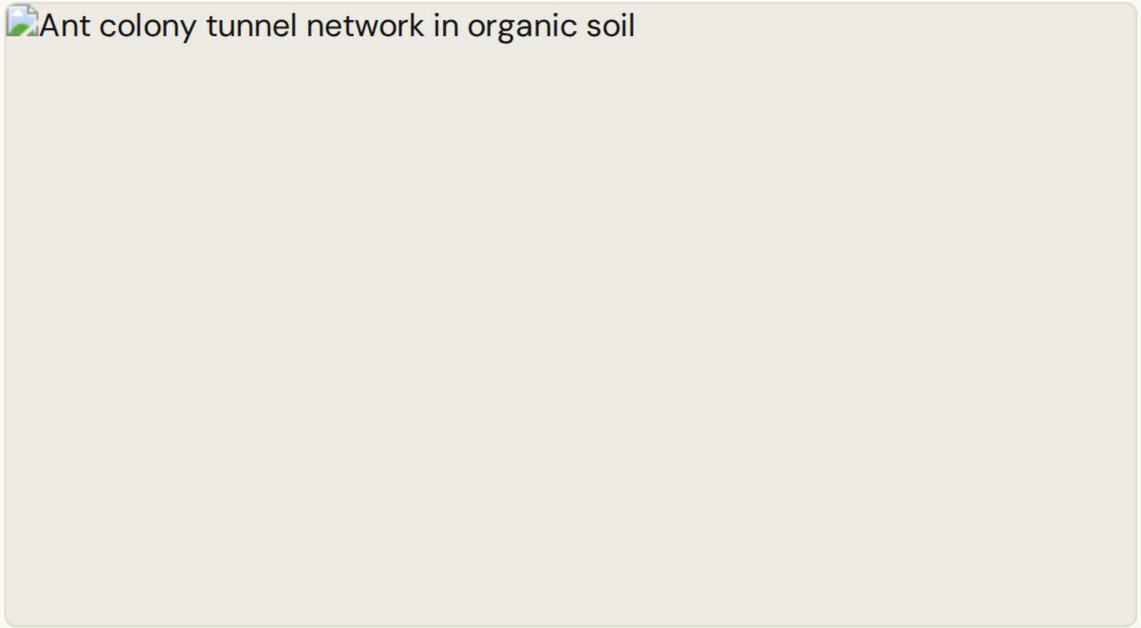
Unlock mutated chemical glands, specialized ant castes, and adaptive biome traits.

SIMULATION DEEP DIVE

Living Ecosystem Sandbox

Everything Interacts:

- **Multi-Species Dynamics:** Predators, prey, rival colonies, and fungal flora.
- **Ecosystem Weather:** Floods wash away pheromone pathways; rain alters scent dispersion.
- **Emergent Hazards:** Fire spreads via soil organic matter, creating high-risk tactical barriers.
- **Zero Scripting:** Stories occur purely because independent agent AI reacts to physics.



MARKET PROOF

Simulation Commercial Benchmarks

TITLE	SALES SCALE	CORE APPEAL	SWARM ALIGNMENT
RimWorld	5M+ Units	Emergent AI Storytelling	High (Autonomous Colony)
Factorio	5M+ Units	System Automation	High (Pheromone Flow)
Oxygen Not Included	Multi-Million	Physics & Resource Simulation	High (Fluid & Smell Diffusion)
Noita	Multi-Million	Simulated Pixel Physics	High (Granular Interactions)

Conclusion: Millions of gamers prefer complex systems creating unexpected organic stories over rigid scripted gameplay.

COMPETITIVE ADVANTAGE

AI as the Gameplay Engine

100%

AUTONOMOUS AI

Systemic Shift in Strategy

Instead of units... You control environmental information.

Instead of scripted AI... AI decisions are the primary gameplay loop.

Instead of traditional RTS... Ecology itself becomes the core strategy.

TECHNICAL ARCHITECTURE

Developer Engineering Solutions

- **Vector-Field Performance Optimization:** High-density pheromone grids calculated via vector fields using C++/Godot GDExtension, keeping frame-times low even with thousands of active agents.
- **Intuitive Indirect Onboarding:** Onboarding players via interactive puzzle scenarios rather than walls of text, teaching players to think in chemical vectors rather than direct clicks.
- **Decoupled Simulation Engine:** Agent decision trees decoupled from environmental diffusion pipelines, allowing simple extension for future insect species.

DEVELOPMENT PLAN

Product Roadmap & Milestones

Stage 1: Prototype

Core ant movement, food harvesting, 4 pheromone types. Goal: Satisfying movement.

Stage 2: Combat

Combat pheromones, predator insects, dynamic rival colonies.

Stage 3: World

Rain, flooding, soil diffusion, dynamic weather hazards.

Stage 4: Steam EA

Steam Next Fest demo, community workshop, procedural sandbox mode.

BUSINESS MODEL

Monetization & Creator Hooks

Monetization Strategy

Base Price: $15–$20 Premium release on Steam (Simulation players prefer complete games).

Expansions: Ecosystem DLCs (Amazon Rainforest, Desert Dunes, Mangroves, Underground).

Modding: Steam Workshop integration for community insect species & maps.

Viral Creator Hook

Emergent games generate massive organic content on YouTube & TikTok:

"My ants accidentally caused an ecological collapse..."

Key Creator Outreach: *Splattercat, Real Civil Engineer, Wanderbots, Aliensrock.*

BEYOND A SINGLE GAME

The Ecosystem Simulation Engine

Building the foundational engine for emergent adaptive systems—extending from termites and bees to slime molds, autonomous AI agents, and research simulations.

swarm-game.com | dev@swarm-game.com

Image Sources



https://images.stockcake.com/public/0/f/0/0f033732-df6a-46f0-a9fd-0b49a707269f_large/ant-colony-cross-section-stockcake.jpg

Source: stockcake.com